



A moving boundary model for food isothermal drying and shrinkage: General setting

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ABSTRACT

A moving-boundary model is proposed for describing food isothermal drying. The model takes into account volume reduction of food materials and it is capable to predict sample shrinkage and surface deformation during the drying process. It can be applied to any sample geometry (discoid, cylindrical, cubic, parallelepiped) and to any food material since it can take into account that sample volume can decrease of a quantity that can be smaller, equal or larger than the corresponding volume of removed water.

The core of the present model is the adoption of a pointwise shrinkage velocity equal (and opposite in sign) to the water diffusive flux times a shrinkage factor $\alpha(\phi)$ depending on the pointwise water volume fraction $\phi(\mathbf{x})$. The shrinkage factor $\alpha(\phi)$ can be assumed a priori or directly derived from experimental data of the rescaled volume V/V_0 vs the rescaled moisture content X/X_0 . For ideal shrinkage $\alpha(\phi) = 1$.

The model provides good results in terms of prediction of volume reduction, surface deformation and effective water diffusivity for potato strips air-drying.

1. Introduction

Drying is one of the most widespread processes for food preservation and storage (Arvanitoyannis et al., 2005; Albanese et al., 2006). Process conditions strongly influence the final results in terms of food quality, which can be often connected to changes in color, dimensions and visual aspect, as well as in terms of energy efficiency (Ansari and Datta, 2003; Raitio et al., 2011). Drying is, in fact, one of the most energy consuming industrial processes (Kudra, 2004).

Different systems configurations have to be adopted to optimize drying processes (e.g. Moejes et al. (2018); El-Mashad and Pan (2017)). In this context, the importance of mathematical modeling is crucial as it can be profitably used, in food process design and control, to analyze a wide range of conditions (Defraeye, 2014).

The use of mathematical models also provides deeper insights into the physics of processes (Llave et al., 2016) and materials (Brasiello et al., 2010, 2012) at different length and time scales as demonstrated by the flourishing research activities. In literature, it is possible to distinguish between empirical, semi-theoretical and theoretical models according to growing complexity. In most cases, the forced choice, due to the lack of knowledge of complex transport phenomena occurring or of details about food matrices, is the adoption of an empirical one (Mayor and Sereno, 2004), consisting in a pure kinetic model for the

evolution of the variables taken into account for the description of the process (Akpınar and Bicer, 2005; Ertekin and Yaldiz, 2004). Empirical models are quite often used for their easy handling in terms of computational costs. The main drawback is the limited predictive capability outside the experimental conditions used for parameters estimation.

In very simple (and few) cases, it is possible to write a theoretical model, based on first principles, in which the relevant physics is accounted for. Despite of high numerical burdens, this represents a very favorable condition in all the cases in which a deeper insight into the process is needed and related experiments are very expensive (e.g. for process scale-up).

Sometimes, the lack of information does not allow their use: however, in such cases, it is possible to modify theoretical models by adding empirical expressions whose parameters have to be determined through experimental sessions (Hassini et al., 2007; Brasiello et al., 2010, 2011). These models are called semi-theoretical and represent a relatively simple and flexible choice in a lot of practical cases.

In the case of food drying, the adoption of semi-theoretical models seems the forced choice since the occurrence of material modifications prevents the adoption of pure theoretical ones. Shrinkage is in fact very common during drying of foods (Pacheco-Aguirre et al., 2015; Brasiello et al., 2011). To take it into account, a deep comprehension of the physics is needed. In several examples, available in literature, shrinkage

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modeling consists in the introduction of a further hypothesis about the effect of volume reduction on the materials and several authors (Torrez Irigoyen et al., 2014; Da Silva et al., 2014) relate the volume reduction to the moisture content. In the paper of Aprajeeta et al. (2015), the hypothesis of constant volume reduction in time is adopted while in Curcio and Aversa (2014), a specific hypothesis on the stress tensor (evolving with the local values of water content) is made. While the last approach introduces deep insights into the physics of the process, it is however very computationally expensive with respect to the first one. Another possibility in order to meet more affordable computational costs is the introduction of either a fictitious convective term (Carslaw and Jaeger, 1959) or (which is the same in most cases see e.g. Brasiello et al. (2013)) a variable model parameter (e.g. the diffusion coefficient) through which considering the effect of shrinkage on water content evolution in time (Ortiz-Garcia-Carrasco et al., 2015; López-Méndez et al., 2018).

In the works of Brasiello et al. (2013, 2017) such approaches are both fruitfully used to model the drying process of eggplants samples with good results in terms of prediction of the evolution of the total water content as well as the water content profiles obtained through Magnetic Resonance Imaging (MRI). However, their approach is limited from the point of view of the prediction of volume reduction in time. The model only allows an overall estimation due to the fixed boundary approximation.

In this paper a theoretical model with moving boundaries is adopted to describe both the water content and volume/surface evolution in time. This approach has the great advantage of capturing all the essential features of the process with a quite simple formulation and can be applied to any sample geometry (discoid, cylindrical, cubic, parallelepiped).

The core of the proposed model is the adoption of a pointwise shrinkage velocity $\mathbf{v}(\mathbf{x}) = -\alpha(\phi)\mathbf{J}(\mathbf{x}) = \alpha(\phi)\mathbf{D}\nabla\phi(\mathbf{x})$ equal (and opposite in sign) to the water diffusive flux $\mathbf{J}(\mathbf{x})$ times a shrinkage factor $\alpha(\phi)$ depending on the pointwise water volume fraction $\phi(\mathbf{x})$. The shrinkage factor can be assumed a priori or can be directly derived from experimental data of the rescaled volume V/V_0 vs the moisture ratio (also called rescaled moisture content) X/X_0 . For ideal shrinkage, where the decrease of the material volume is exactly equal to the volume of the removed water, $\alpha(\phi) = 1$.

The introduction of the pointwise shrinkage velocity proportional to the water diffusive flux $\mathbf{J}$ marks the difference between the proposed model and other literature moving boundary models (mainly 1-d models) accounting for sample thickness variation as a function of time $L(t)$, e.g. Hassini et al. (2007), but still relying on the direct experimental determination of $L(t)$ because the model cannot predict the time evolution of the sample thickness even in the simple case of ideal shrinkage.

In order to test the prediction capability of the moving boundary model, a set of literature experimental data resulting from the air-drying process of potato strips is investigated (Section 5), thus proposing a fully analytical approach to determine the optimal shrinkage factor. The model provides good results in terms of prediction of volume reduction, surface deformation and effective water diffusivity.

2. Mathematical model

The temporal evolution of the water content during isothermal drying of a food matrix (pulp and water) with volume $V(t)$ and boundary $S(t)$, both evolving in time, can be described by a system of partial differential equations.

Although food matrices are in general very complex, and therefore very far from being homogeneous and isotropic, these two assumptions are often both enforced, as in the present case, in order to streamline the model.

Let $\mathbf{v}(\mathbf{x}, t)$ be the pointwise shrinkage velocity. It is assumed that the local mass flux of water, at every point $\mathbf{x}$ of the domain, is purely

diffusive and equal to $\mathbf{j}$ if observed from a reference frame with the origin at $\mathbf{x}$ and moving with velocity $\mathbf{v}$ (Tu and Ouano, 1977). Hence, the mass balance, in a fixed reference frame with respect to the laboratory, consists of the following equation:

$$\frac{\partial c(\mathbf{x}, t)}{\partial t} + \nabla \cdot (c\mathbf{v}) = -\nabla \cdot \mathbf{j}, \quad \mathbf{x} \in V(t) \quad (1)$$

where $c(\mathbf{x}, t)$ is the mass concentration of water and $\nabla \cdot (c\mathbf{v})$ is a convective term arising from local shrinkage.

Eq. (1) can be rewritten in terms of volume fraction $\phi(\mathbf{x}, t) = c(\mathbf{x}, t)/\rho_w$, under the hypothesis of constant water density (ρ_w):

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi\mathbf{v}) = -\nabla \cdot \mathbf{J}, \quad \mathbf{x} \in V(t) \quad (2)$$

in which $\mathbf{J} = \mathbf{j}/\rho_w$.

Introducing the Fick's first law of diffusion $\mathbf{j} = -D\nabla c$ or equivalently $\mathbf{J} = -D\nabla\phi$, Eq. (2) becomes:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi\mathbf{v}) = \nabla \cdot (D\nabla\phi), \quad \mathbf{x} \in V(t) \quad (3)$$

D being the diffusion coefficient.

A functional form for $\mathbf{v}$ together with an equation describing the evolution of the boundary are needed.

The macroscopic balance over the volume $V(t)$, obtained by integration of both sides of Eq. (3), provides all the functional forms for $\mathbf{v}$ consistent with the hypotheses:

$$\int_{V(t)} \frac{\partial \phi}{\partial t} dV = \int_{V(t)} \nabla \cdot (D\nabla\phi) dV - \int_{V(t)} \nabla \cdot (\phi\mathbf{v}) dV \quad (4)$$

which, by applying both Leibniz integral rule and Gauss Theorem, becomes:

$$\frac{d}{dt} \int_{V(t)} \phi dV = \oint_{S(t)} (D\nabla\phi) \cdot \mathbf{n} dS - \oint_{S(t)} \phi(\mathbf{v} - \mathbf{v}_S) \cdot \mathbf{n} dS \quad (5)$$

in which the boundary velocity $\mathbf{v}_S = \mathbf{v}|_{\mathbf{x}_b}$ is introduced, i.e. the shrinkage velocity at any point $\mathbf{x}_b \in S(t)$ on the boundary $S(t)$.

The hypothesis of purely diffusive local flow implies that the second term at the right-hand side of Eq. (5) is null, because only in this case the total amount of water leaving the volume V per unit time equals the integral of diffusive flux over the boundary S . Therefore, although any functional form of $\mathbf{v}$ can be chosen, in principle, in order to fulfill the macroscopic balance Eq. (4), the boundary velocity has to satisfy the equation:

$$\mathbf{v} = \mathbf{v}|_{\mathbf{x}_b} \quad (6)$$

A constitutive equation for $\mathbf{v}$, specific for the class of materials under investigation, is thus needed both in Eq. (3) and in Eq. (6).

It can be observed that Eq. (3) and Eq. (6) describing the shrinking process of food samples is exactly the same set of equations describing the swelling of rubbery polymers (Papanu et al., 1989; Adrover and Nobili, 2015) by simply identifying the pointwise shrinkage velocity $\mathbf{v}$ with the pointwise swelling velocity $\mathbf{v}_{sw}$.

In a drying process, food is releasing water, this resulting in a global shrinkage process, i.e. a movement of the boundary in the direction of the water concentration gradient. In a swelling process, the polymer matrix is absorbing the solvent, this resulting in a global polymer network swelling, i.e. a movement of the boundary in the direction of the solvent concentration gradient. Hence, what distinguishes between swelling and shrinking is the sign of the concentration gradient.

For the swelling process, since the solvent and the polymer are assumed to be incompressible and to mix with no volume change, the local swelling velocity is $\mathbf{v}_{sw} = -\mathbf{J}$ and the polymer matrix volume increases exactly of the same volume of solvent absorbed by the system.

In the case of drying process of food matrices, the physics is more complex and experimental observations suggest that sample volume

can decrease of a quantity that can be smaller, equal or larger than the corresponding volume of released water, i.e.

$$\frac{dV}{dt} \leq \frac{1}{\rho_w} \frac{dM_w}{dt} \quad (7)$$

in which M_w is the mass of water, still present in the sample at time t .

The symbol $<$ holds, for instance, when dealing with very rigid matrices able to partially preserve their initial structures, while the symbol $>$ holds when the water flow induces deep structural modifications of the pulp matrices (pulp compressibility). The symbol $=$ holds for ideal shrinkage.

For this reason, still pursuing the analogy between food shrinkage and polymer swelling, the following very general constitutive equation for $\mathbf{v}$ is adopted:

$$\mathbf{v} = -\alpha(\phi)\mathbf{J} = \alpha(\phi) D \nabla \phi \quad (8)$$

implying that the pointwise shrinkage velocity $\mathbf{v}$ is proportional to the water concentration gradient times a proportionality function $\alpha(\phi)$ tuning, at each point of the system, the relationship between water flux $\mathbf{J}$ and volume reduction. The shrinkage factor $\alpha(\phi)$ is, in principle, a pointwise function as it is assumed to be dependent on the local water volume fraction ϕ . With the introduction of Eq. (8), the three conditions for Eq. (7) can be easily achieved through a suitable choice of the function $\alpha(\phi)$. The case of ideal shrinkage corresponds to $\alpha(\phi) = 1$.

In Section 3 admissible functions for $\alpha(\phi)$ are investigated as well as the influence of different functional forms for $\alpha(\phi)$ on the drying process. Section 5, focusing on a specific food drying process, focuses on how to get information on the functional form of $\alpha(\phi)$ directly from experimental data of rescaled volume and moisture ratio.

The simplest case is that of a constant value, i.e. $\alpha(\phi) = C$. The lower limit is $C = 0$ in the case of a fully rigid solid and therefore a fixed boundary model system. $C = 1$ represents the case that volume reduction corresponds exactly to the volume of water flowing outside the system. Values of C less or greater than unity imply volume reduction less or greater than the corresponding water volume flow, respectively. It must be observed that, assuming $\alpha(\phi) = C$ does not imply that the shrinkage velocity is constant but that, according to Eq. (8), the shrinkage velocity is directly proportional to the concentration gradient. Therefore the shrinkage velocity asymptotically tends to zero at every point when equilibrium conditions are reached and the water concentration gradient is null everywhere in the system.

By substituting Eq. (8) into Eqs. (3) and (6), one obtains:

$$\frac{\partial \phi}{\partial t} = -\nabla \cdot (\phi (\alpha D \nabla \phi)) + \nabla \cdot (D \nabla \phi) = \nabla \cdot (D \nabla \phi (1 - \alpha \phi)), \quad \mathbf{x} \in V(t) \quad (9)$$

$$\frac{d\mathbf{x}_b}{dt} = \mathbf{v}_s = \alpha D \nabla \phi|_{\mathbf{x}_b}, \quad \mathbf{x}_b \in S(t) \quad (10)$$

which completely define the moving boundary problem together with the following mixed (Robin) boundary condition for ϕ at each point $\mathbf{x}_b \in S(t)$, defining the flowing water from the boundary of the system towards the environment:

$$-D \nabla \phi \cdot \mathbf{n}|_{\mathbf{x}_b} = h_m \frac{\rho_{air}}{\rho_w} (Y|_{\mathbf{x}_b} - Y_{eq}) = h_m K_p \frac{\rho_{air}}{\rho_s} (\phi|_{\mathbf{x}_b} - \phi_{eq}) \quad (11)$$

in which $\mathbf{n}$ is the outward-pointing normal unit vector, h_m is a mass transfer coefficient [m/s], Y is the air moisture content [kg water/kg dry air], ρ_{air} is the air density on dry basis [kg dry air/m³], ρ_s is the solid (pulp) density [kg pulp/m³product] and K_p is the water partition ratio between the gas and the solid phases $Y = K_p c/\rho_s = K_p (\rho_w/\rho_s) \phi$. The subscript eq stands for equilibrium values. For simplifying the notation, the explicit dependence of α on ϕ is dropped in Eqs. (9) and (10), thus simply writing $\alpha(\phi) = \alpha$.

The water diffusion coefficient D can be assumed constant or, in a more refined approach, it can be considered an increasing exponential

function of the water volume fraction (as usually adopted in the swelling literature)

$$D(\phi) = D_0 \exp\left(-\beta \frac{\phi_0 - \phi}{\phi_0 - \phi_{eq}}\right), \quad \beta \geq 0 \quad (12)$$

where ϕ_0 is the initial water volume fraction (supposed to be uniform all over the sample), D_0 and $D_0 \exp(-\beta)$ are the effective diffusivities at the beginning and at the end of the drying process, respectively.

In the rest of the paper constant water diffusivity is assumed i.e. $\beta \simeq 0$, $D = D_0$.

By introducing the dimensionless space and time variables $\tau = tD/L_r^2$, $\tilde{\mathbf{x}} = \mathbf{x}/L_r$, $\tilde{V} = V/L_r^3$, $\tilde{S} = S/L_r^2$ and the dimensionless differential operators $\tilde{\nabla} = \nabla/L_r$, $\tilde{\nabla} \cdot = \frac{\nabla}{L_r} \cdot$, L_r being a characteristic reference length, the moving boundary model equations for the normalized water volume fraction $\psi = \phi/\phi_0$ attain the form:

$$\frac{\partial \psi}{\partial \tau} = \tilde{\nabla} \cdot (\tilde{\nabla} \psi (1 - \alpha \psi)), \quad \tilde{\mathbf{x}} \in \tilde{V}(\tau) \quad (13)$$

$$-\tilde{\nabla} \psi \cdot \mathbf{n}|_{\tilde{\mathbf{x}}_b} = Bi_m (\psi|_{\tilde{\mathbf{x}}_b} - \psi_{eq}), \quad \tilde{\mathbf{x}}_b \in \tilde{S}(\tau) \quad (14)$$

$$\frac{d\tilde{\mathbf{x}}_b}{d\tau} = \alpha \phi_0 \tilde{\nabla} \psi|_{\tilde{\mathbf{x}}_b}, \quad \tilde{\mathbf{x}}_b \in \tilde{S}(\tau) \quad (15)$$

where Bi_m is the mass transfer Biot number $Bi_m = \frac{HL_r}{D} = \frac{h_m L_r}{D} K_p \frac{\rho_{air}}{\rho_s}$. Initial conditions are $\tilde{V}(0) = \tilde{V}_0$, $\tilde{S}(0) = \tilde{S}_0$ and $\psi(\tilde{\mathbf{x}}, 0) = 1$ assuming initial uniform water distribution.

For the 2-d Cartesian problem analyzed in sections 3 and 5 the initial sample domain is a square $-L_0/2 \leq x \leq L_0/2$, $-L_0/2 \leq y \leq L_0/2$, L_0 being the side length. Therefore, L_0 is chosen as the reference length $L_r = L_0$, and the dimensionless space variables $(\xi, \eta) = (x/L_0, y/L_0)$ and the dimensionless operators

$$\tilde{\nabla} f = \frac{\partial f}{\partial \xi} \mathbf{e}_\xi + \frac{\partial f}{\partial \eta} \mathbf{e}_\eta, \quad \tilde{\nabla} \cdot \mathbf{f} = \frac{\partial f_\xi}{\partial \xi} + \frac{\partial f_\eta}{\partial \eta} \quad (16)$$

are introduced. The dimensionless initial domain is $(\xi, \eta) \in [-1/2, 1/2] \times [-1/2, 1/2]$.

For the 2-d axial symmetric cylindrical problem analyzed in sections 3 the initial sample domain is a cylinder $-L_z/2 \leq z \leq L_z/2$, $0 \leq r \leq R_0$ with $L_z > R_0$. Therefore, R_0 is chosen as the reference length $L_r = R_0$, and the dimensionless space variables $(\rho, \zeta) = (r/R_0, z/R_0)$ and the dimensionless operators

$$\tilde{\nabla} f = \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{\partial f}{\partial \zeta} \mathbf{e}_\zeta, \quad \tilde{\nabla} \cdot \mathbf{f} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho f_\rho) + \frac{\partial f_\zeta}{\partial \zeta} \quad (17)$$

are introduced. The dimensionless initial domain is $(\rho, \zeta) \in [0, 1] \times [-a_r/2, a_r/2]$, $a_r = L_z/R_0$ being the cylinder aspect ratio.

The 2-d moving boundary problems have been numerically solved using finite elements method (FEM) in Comsol Multiphysics 3.5.

The real advantage of Comsol Multiphysics is that this FEM commercial software has been originally designed to solve any PDE or set of coupled PDEs and therefore it can be used to numerically solve any well posed transport model. Moreover Comsol is perfectly suited for handling moving boundary models. Infact with the Moving Mesh (ALE) application mode you can create models where the geometry (actually the mesh) changes shape due to some physics in the model. It possible to study both static and time-dependent deformations where the geometry changes its shape due to the dynamics of the problem.

The convection-diffusion package (conservative form) has been coupled with ALE (Arbitrary Lagrangian Eulerian) moving mesh, allowing remeshing during the time evolution of the physical domain. Free displacement induced by boundary velocity conditions has been set. Lagrangian quadratic elements have been chosen. The linear solver adopted is UMFPACK, with relative tolerance 10^{-3} and absolute tolerance 10^{-6} . The number of finite elements has been varied in the range $2 \times 10^5 - 5 \times 10^5$ with a non-uniform mesh. Smaller elements

(maximum element size 5×10^{-3} dimensionless units) have been located close to the moving boundaries in order to accurately compute concentration gradients controlling the velocity of the moving fronts. The final mesh adopted is the results of a large number of numerical simulations performed in order to optimize the mesh and to guarantee the accuracy of numerical results, i.e. that macroscopic results are actually independent of the mesh.

3. Influence of $\alpha(\psi)$ and Bi_m on drying processes

From Eqs. (9) and (10) it can be observed that the role of the pointwise shrinkage velocity $\mathbf{v}$ is twofold. In Eq. (10), it determines a global sample shrinkage through the boundary condition, thus reducing the diffusive paths and enhancing the diffusion process while, on the other hand, it tends to slow down the diffusive process in Eq. (9) through a convective flux $\mathbf{v}\phi$ which is opposite in sign to the diffusive one $\mathbf{J}$. Moreover, the balance equation Eq. (9) highlights that the effect of shrinkage is to modify the diffusion process inside the system, since the actual summation of the convective and diffusive fluxes results in a modified diffusion term with an effective diffusivity $D(1 - \alpha(\psi)\phi_0\psi)$ written in terms of the dimensionless variable ψ (see Brasiello et al. (2013)).

The functional form of the shrinkage factor $\alpha(\psi)$ is thus extremely important in modeling drying experiments. It must be chosen in such a way to guarantee the positiveness of the effective diffusivity for any ψ , and therefore

$$1 - \alpha(\psi)\phi_0\psi > 0 \rightarrow \alpha(\psi) < (\phi_0\psi)^{-1}, \forall \psi \in [\psi_{eq}, 1] \quad (18)$$

The curve $(\phi_0\psi)^{-1}$ defines an upper limit for $\alpha(\psi)$ that is independent of the water diffusivity D , the Biot number Bi_m and the geometry of the system.

Fig. 1 shows the influence of different admissible α functions (shown in the inset) on the behaviour of the drying curves V/V_0 vs X/X_0 for a cylindrical sample characterized by the aspect ratio $a_r = 2.75$ and $\phi_0 = 0.5$. Continuous lines and dash dot lines show the behaviours for $Bi_m = 1$ and $Bi_m = 5$, respectively.

By adopting a standard notation, V/V_0 indicates the rescaled volume and X/X_0 is the moisture ratio (or rescaled moisture content)

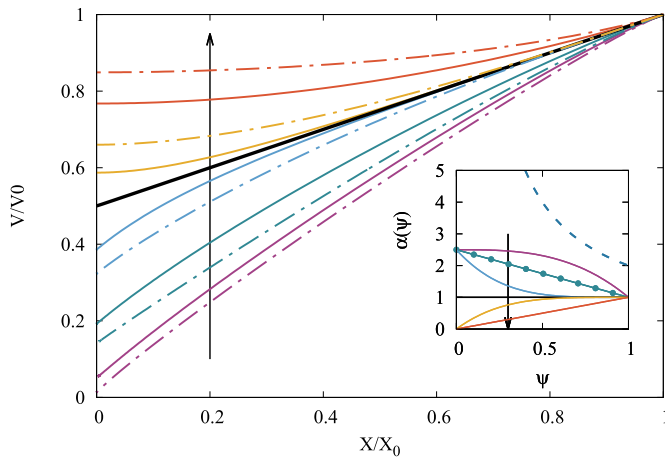


Fig. 1. Influence of $\alpha(\psi)$ on the behaviour of V/V_0 vs X/X_0 for $Bi_m = 1.5$. The initial sample shape is a cylinder with aspect ratio $a_r = 2.75$. $\phi_0 = 0.5$, $\psi_{eq} = 0$. Different curves correspond to different α functions (shown in the inset). Continuous lines correspond to $Bi_m = 1$. Dot-dashed curves correspond to $Bi_m = 5$. Thick black line shows the linear behaviour $V/V_0 = 1 - C\phi_0(1 - X/X_0)$, independent of Bi_m , associated with a constant function $\alpha(\psi) = C$ for $C = 1$. Dashed curve in the inset shows the limiting curve $\alpha(\psi) = (\psi\phi_0)^{-1}$, Eq. (18). Different curves from bottom to top (see arrow in the larger plot) correspond to $\alpha(\psi) = 1 + (3/2)(1 - \psi^3)$, $\alpha(\psi) = 1 + (3/2)(1 - \psi)$, $\alpha(\psi) = 1 + (3/2)(1 - \psi)^4$, $\alpha(\psi) = 1$, $\alpha(\psi) = 1 - (1 - \psi)^4$, $\alpha(\psi) = -\psi$.

$X/X_0 = M_w/M_w^0 = (W - W_d)/(W_0 - W_d)$ where $M_w^0 = \rho_w V_0 \phi_0$ is the initial amount of water in the sample, W is the sample weight (water + pulp) and W_d is the dry sample weight. In terms of the dimensionless model variables the rescaled volume is $V/V_0 = \tilde{V}/\tilde{V}_0$ and the moisture ratio $X/X_0 = \int_{\tilde{V}} \psi(\tilde{\mathbf{x}}) d\tilde{V}$.

By adopting non-constant α functions it is possible to obtain curved shapes for the functions V/V_0 vs X/X_0 which are similar to those frequently observed in experimental data. Curves V/V_0 vs X/X_0 associated with $\alpha(\psi) \geq 1$ lie all below the thick black line representing the linear behaviour $V/V_0 = 1 - C\phi_0(1 - X/X_0)$, while curves V/V_0 vs X/X_0 associated with $\alpha(\psi) \leq 1$ lie all above that line. The thick black line is independent of Bi_m and associated with a constant function $\alpha = C = 1$, representing the ideal shrinkage when volume reduction corresponds exactly to the volume of removed water (see Section 4). It actually represents a simple confirmation of the accuracy of the numerical scheme adopted for solving the moving boundary model.

For the same α function, the curvature effect is slightly more pronounced for smaller values of Bi_m , while Bi_m influences more significantly the asymptotic rescaled volume V_{eq}/V_0 , i.e. V/V_0 for $X/X_0 = X_{eq}/X_0$.

The Biot number Bi_m strongly influences the evolution of the sample shape during the drying process, more significantly the sample shape on long time scales corresponding to low values of the moisture ratio X/X_0 .

Fig. 2 shows spatial profiles for ψ on the evolved sample shape for decreasing values of $X/X_0 = 0.8, 0.2, 0.1$ (from left to right, respectively). Top and bottom panels refer to $Bi_m = 1$ and $Bi_m = 5$, respectively. For both panels, $\alpha(\psi) = C = 1$. It can be observed that lower Bi_m values lead to a smoother and flat boundary profile, while higher values lead to pronounced cusps.

The influence of Bi_m on the sample shape evolution can be quantified by comparing the actual sample surface with that of a sample having the same volume but still keeping the original shape. As an example let us consider a parallelepiped, square section $L_0 \times L_0$ and height $L_z \gg L_0$. This, for instance, is the case of potato sticks analyzed in Section 5. In this case the drying process can be reasonably addressed as a 2-d Cartesian problem by following the time evolution of the water content and the shape of a cross-section of the sample, suitably far from both top and bottom surfaces in order to avoid boundary effects.

Fig. 3 shows water concentration profiles (ψ) on the evolved cross sections for $\alpha(\psi) = C = 1$ and $\phi_0 = 0.5$. Three different values of $Bi_m = 1, 3, 5$, at the same low value of the moisture ratio $X/X_0 = 0.1$ are reported. The three samples at the three different Bi_m exhibit the same (rescaled) surface area $A/A_0 = 1 - C\phi_0(1 - X/X_0) = 0.55$, but very different (rescaled) perimeters P/P_0 , increasing as the value of Bi_m is increased.

Fig. 4 shows the behaviour of P/P_0 vs A/A_0 for $Bi_m = 1, 3, 5$. The dashed curve $(P/P_0)_{sq} = (A/A_0)^{1/2}$ is also reported. It represents the relationship between the rescaled perimeter P/P_0 and the rescaled surface area A/A_0 of a sample that shrinks always keeping the initial square shape.

The larger is Bi_m , the larger is the perimeter and hence the ratio P over A . In a more general 3-d case, the larger is Bi_m , the larger is the ratio of surface area over sample volume because of the significant shape deformation and of the surface stretching at the edges.

These observation are extremely useful because the estimation of the Bi_m value can be realized via experiments, starting from the simple visual information of the final shape of the sample subjected to the drying process as showed, for instance, in the case of potato strips (see Fig. 9) analyzed in Section 5.

4. Evaluation of the shrinkage factor $\alpha(\psi)$ from experimental data

In order to estimate the shrinkage factor $\alpha(\psi)$, it is possible to rely on the experimental data V/V_0 vs X/X_0 (see Madiouli et al. (2012), Lozano et al. (1983) and references therein).

From the macroscopic material balance Eq. (5), by enforcing the

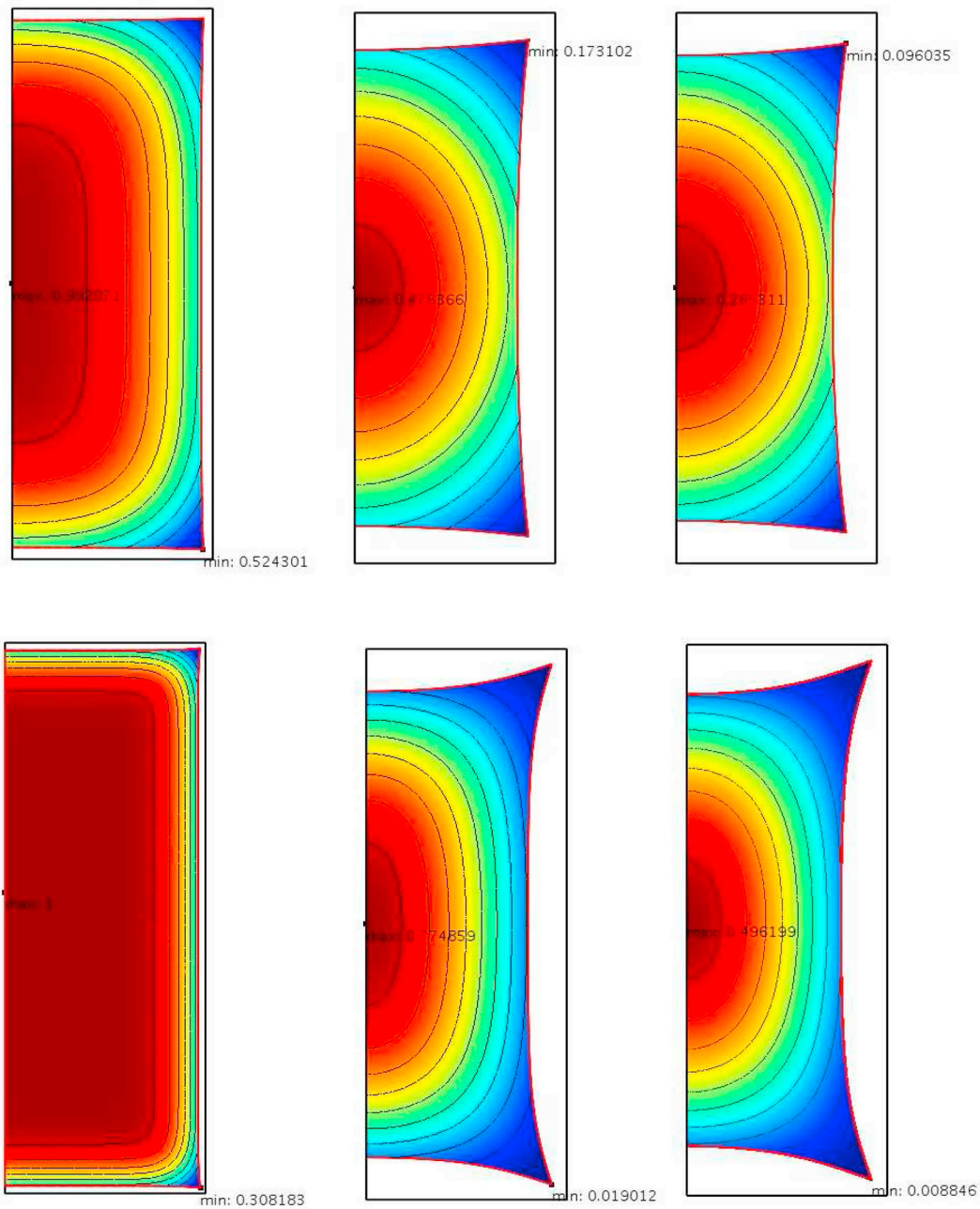


Fig. 2. Contour plot of ψ on the evolved shape of the half cylinder during the drying process for $X/X_0 = 0.8, 0.2, 0.1$ (from left to right). $\alpha(\psi) = 1, \phi_0 = 0.5$. Top panel $Bi_m = 1$. Bottom panel $Bi_m = 5$.

consistency equation $\mathbf{v}_S = \mathbf{v}|_S$ and the constitutive equation $\mathbf{v} = \alpha(\phi)D\nabla\phi$ we obtain

$$\frac{dM_w}{dt} = \oint_{S(t)} \rho_w \frac{\mathbf{v} \cdot \mathbf{n}}{\alpha(\psi)} dS \simeq \frac{\rho_w}{\alpha(\psi_b)} \oint_{S(t)} \mathbf{v} \cdot \mathbf{n} dS = \frac{\rho_w}{\alpha(\psi_b)} \frac{dV}{dt} \quad (19)$$

where M_w is the amount of water in the sample at time t and ψ_b is the normalized water volume fraction evaluated, at time t , in a suitable point, called probe point P , on the sample surface. The probe point position evolves in time according with the sample surface on which it is initially attached. ψ_b is clearly a function of time and can be related to the evolution in time of M_w . In the following, the choice of a probe point characterized by the maximum displacement (i.e. a point characterized by the maximum shrinkage) is adopted for ψ_b .

Only in the case of $\alpha(\psi) = C$, Eq. (19) is an exact equation and can be further integrated over time, obtaining:

$$M_w - M_w^0 = (\rho_w/C)(V - V_0) \quad (20)$$

Eq. (20) can be conveniently rewritten in terms of the moisture ratio X/X_0 or in terms of the rescaled volume of removed water V_{rw}/V_0

$$\frac{V_{rw}}{V_0} = \frac{M_w^0 - M_w}{\rho_w V_0} = \phi_0 \left(1 - \frac{X}{X_0} \right) \quad (21)$$

as follows:

$$1 - V/V_0 = C \phi_0 \left(1 - \frac{X}{X_0} \right) = C V_{rw}/V_0 \quad (22)$$

From Eq. (22) is evident that for $C = 1$, the volume reduction equals, at each time instant, the volume of removed water.

In the more general case of non-constant $\alpha(\psi)$, Equation (19), written in terms of the moisture ratio X/X_0 , reads as

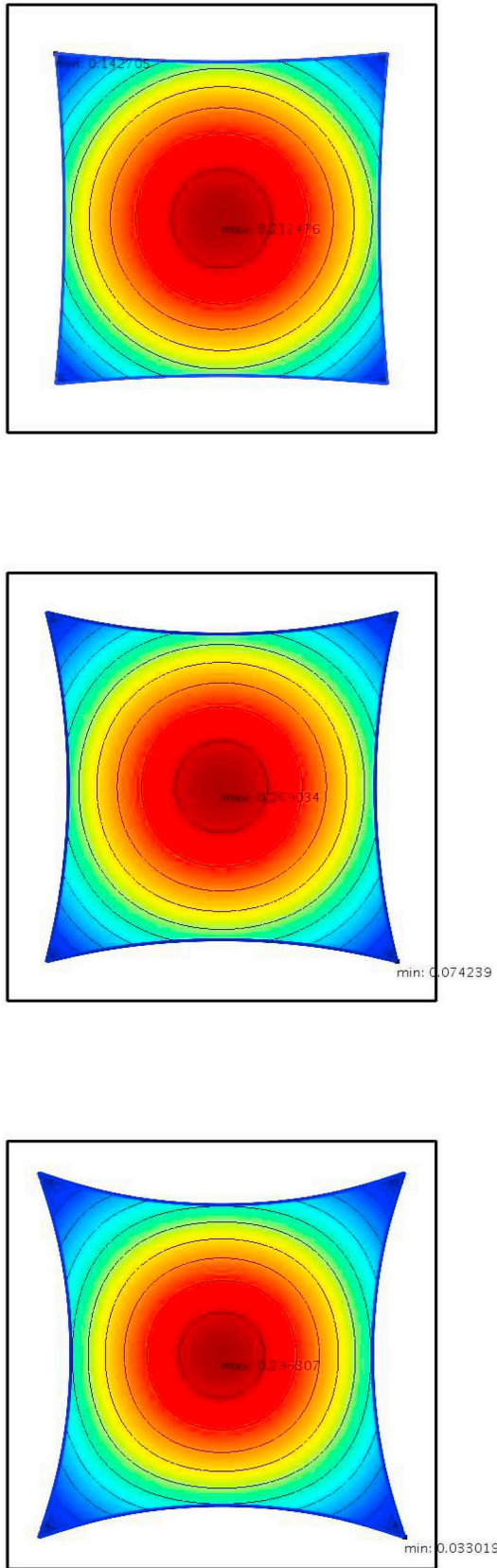


Fig. 3. Contour plot of ψ on the evolved shape of the initially square sample for $X/X_0 = 0.1$, $A/A_0 = 0.55$. $\alpha(\psi) = 1$, $\phi_0 = 0.5$. From top to bottom $Bi_m = 1$, $Bi_m = 3$, $Bi_m = 5$.

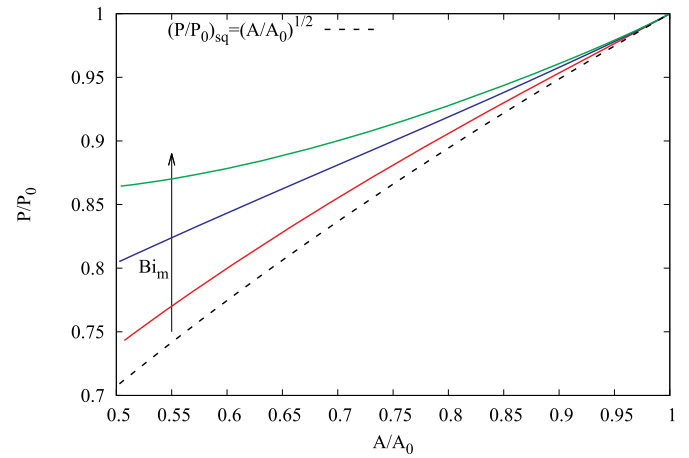


Fig. 4. P/P_0 vs A/A_0 for $\alpha(\psi) = 1$, $\phi_0 = 0.5$, $Bi_m = 1, 3, 5$. Arrow indicates increasing values of Bi_m . Dashed line shows the behaviour $(P/P_0)_{sq} = (A/A_0)^{1/2}$.

$$\frac{d(X/X_0)}{dt} \simeq \frac{1}{\phi_0 \alpha(\psi_b)} \frac{d(V/V_0)}{dt} \rightarrow \alpha(\psi_b) \simeq \frac{1}{\phi_0} \frac{d(V/V_0)}{d(X/X_0)} \quad (23)$$

Let $G(X/X_0)$ be the experimental curve $V/V_0 = G(X/X_0)$ and $G'(X/X_0) = d(V/V_0)/d(X/X_0)$ its derivative with respect to X/X_0 .

The derivative $G'(X/X_0)$ furnishes an experimentally derived shrinkage factor $\tilde{\alpha}(X/X_0)$

$$\tilde{\alpha}(X/X_0) = \frac{1}{\phi_0} G'(X/X_0) \quad (24)$$

that depends on the integral quantity X/X_0 and not on the pointwise concentration ψ_b .

Therefore, in order to obtain the shrinkage factor $\alpha(\psi_b)$ one needs to relate X/X_0 to ψ_b or, in other words, to introduce a function $X/X_0 = g(\psi_b; Bi_m)$ so that

$$X/X_0 = g(\psi_b; Bi_m) \rightarrow \alpha(\psi_b) = \tilde{\alpha}(g(\psi_b; Bi_m)) = \frac{1}{\phi_0} G'(g(\psi_b; Bi_m)) \quad (25)$$

It is worth noting that the proper choice of the function g is deeply influenced by the choice of the probe point for ψ_b since the function g relates the integral quantity X/X_0 to the values of ψ_b which is picked out in order to be representative of ψ over the shrinking surface.

Once the expression for $\alpha(\psi_b)$ is derived, since ψ_b explores the entire range of values of water volume fraction (from $\psi_b = 1$ at the beginning of the drying process to $\psi_b = \psi_{eq}$ at the end of the process), it is reasonable to adopt the same expression for $\alpha(\psi)$ to evaluate the shrinkage velocity at each point in the sample domain.

The experimental curve $G(X/X_0)$ can be easily obtained through approximation of experimental data V/V_0 vs X/X_0 by means of a least-square best fit with a polynomial function $G(X/X_0) = 1 - \sum_{j=1}^n \beta_j (1 - X/X_0)^j$.

In Section 5 it is shown how to determine the shrinkage factor $\alpha(\psi)$ from $\tilde{\alpha}(X/X_0)$ and apply the moving boundary model to literature experimental data resulting from the air-drying process of potato strips.

5. Potato strips air-drying

In this section, the moving-boundary model is applied to the qualitative and quantitative description of experimental data of potato strips air-drying as reported by [Ortiz-García-Carrasco et al. \(2015\)](#). These authors adopt digital image analysis to estimate the simultaneous shrinkage and shape change during food drying.

Potatoes were washed, dried with a cloth and sliced with a vegetable chipper ($L_0 = 9.525$ mm-square openings) to produce strips that were further cut to their desired length ($L_z = 80$ mm). The initial moisture content was 84.5 ± 3.1 g water/100 g product. The initial water volume fraction was $\phi_0 = 0.43$ (from USDA US Department of Agriculture (May 2016)).

Samples were subjected to convective drying at $T = 50, 60, 70, 80^\circ\text{C}$ for about 330 – 470 min with an air velocity of 2 m/s.

Available experimental data are: (1) the dimensionless moisture ratio X/X_0 vs time; (2) the rescaled cross-sectional area A/A_0 and the rescaled perimeter P/P_0 vs the moisture ratio X/X_0 for slices (1 mm thickness) cut in the middle of potato strips; (3) edge deformation of slices at increasing time instants during the drying process.

By considering the high value of the aspect ratio $L_z/L_0 \simeq 8.4$ and the high value of the surface ratio $S_l/(S_t + S_b) \simeq 16.8$ where $S_l = 3048$ mm² is the lateral surface area and $S_t = S_b = 90.725$ mm² are the top and bottom surface areas, it is reasonable to neglect the drying occurring at the top and bottom strip surfaces and to neglect shrinkage along the longitudinal direction. It can be therefore adopted a two-dimensional model in Cartesian coordinates describing the time evolution of the moisture content, the cross-section area and perimeter of the thin slice.

For this 2-d system (actually a 3-d system invariant along the longitudinal coordinate $0 \leq z \leq L_z$), Eq. (23) reads as

$$\frac{d(X/X_0)}{dt} = \frac{1}{\phi_0 \alpha(\psi_b)} \frac{d(A/A_0)}{dt} \quad (26)$$

$$\tilde{\alpha}(X/X_0) = \frac{1}{\phi_0} \frac{d(A/A_0)}{d(X/X_0)} = \frac{1}{\phi_0} G'(X/X_0) \quad (27)$$

$$A/A_0 = G(X/X_0) \quad (28)$$

where ψ_b is the normalized water volume fraction at the probe point P_1 . It is chosen P_1 as the point of initial coordinates $(\xi, \eta) = (0, 1/2)$ because its displacement is a direct measure of the global shrinkage of the sample along one axis. For symmetry reasons equivalent points are $(\xi, \eta) = (0, -1/2), (1/2, 0), (-1/2, 0)$.

Fig. 5 A shows the behaviour of A/A_0 vs X/X_0 and the comparison with the least-square best fit quadratic curve

$$G(X/X_0) = 1 - \beta_1 \left(1 - \frac{X}{X_0}\right) - \beta_2 \left(1 - \frac{X}{X_0}\right)^2 \quad (29)$$

with $\beta_1 = 0.387$ and $\beta_2 = 0.444$, from which:

$$\tilde{\alpha}(X/X_0) = \left(\frac{\beta_1}{\phi_0}\right) + 2\left(\frac{\beta_2}{\phi_0}\right)\left(1 - \frac{X}{X_0}\right) \quad (30)$$

The experimental shrinkage factor $\tilde{\alpha}(X/X_0)$ increases during the

drying process, i.e. it is a linear decreasing function of X/X_0 almost independent of the drying temperature. The volume reduction is significantly larger than the volume of the removed water as can be observed by the comparison of experimental data with the curve $A/A_0 = 1 - \phi_0 C(1 - X/X_0)$, i.e. Eq. (22) with $C = 1$.

Fig. 5 B shows the behaviour of P/P_0 vs X/X_0 and the comparison with $(P/P_0)_{sq} = (A/A_0)^{1/2} = (G(X/X_0))^{1/2}$ representing the rescaled perimeter of a sample having the same surface area but still keeping the square shape. For small values of the moisture ratio X/X_0 the experimental values for P/P_0 are significantly larger than $(P/P_0)_{sq}$, thus implying that the sample is changing its shape during the drying process, from a square towards a different shape characterized by a larger perimeter/area ratio.

This observation, together with the preliminary considerations made in Section 3 on the influence of Bi_m on the perimeter/area ratio, strongly suggests to adopt a large value for Bi_m , namely $Bi_m \geq 5$ for an accurate prediction of these experimental data.

5.1. Estimation of $\alpha(\psi)$ from $\tilde{\alpha}(X/X_0)$

Once the experimental shrinkage factor $\tilde{\alpha}(X/X_0)$ has been evaluated through Eq. (30), it is necessary to relate X/X_0 to ψ_b through a function in order to obtain $\alpha(\psi)$ (see Eq. (25)).

As a first shortcut model it can be assumed

$$X/X_0 = g(\psi_b; Bi_m) = \psi_b \quad (31)$$

Obviously this is a very rough model, as it is actually independent of Bi_m . However, it represents a quite good compromise between simplicity and the need of an adequate physical description of a complex process.

The limit of the functional choice made in Eq. (31) can be determined by observing that X/X_0 can be rewritten in terms of the average rescaled water volume fraction ψ_{av} (averaged over the entire sample volume)

$$X/X_0 = \psi_{av} V/V_0 = \psi_{av} A/A_0 \quad (32)$$

and that $\psi_{av} \geq \psi_b$ at any time during the drying process and for any Bi_m . Therefore, it is expected that, at the beginning of the drying process, when A/A_0 (and hence V/V_0) is still order of unity, $X/X_0 \simeq \psi_{av} > \psi_b$. On the contrary, at larger times, when ψ_{av} is slightly larger than ψ_b and the volume reduction becomes important $A/A_0 = V/V_0 \ll 1$, it is expected that $X/X_0 \simeq \psi_b A/A_0 < \psi_b$ (see numerical results shown in Appendix supporting this observation).

Hence, the linear model Eq. (31) underestimates the moisture ratio X/X_0 at the beginning of the process but, at the end, it tends to overestimate it. The magnitude of these effects has to be evaluated on a case

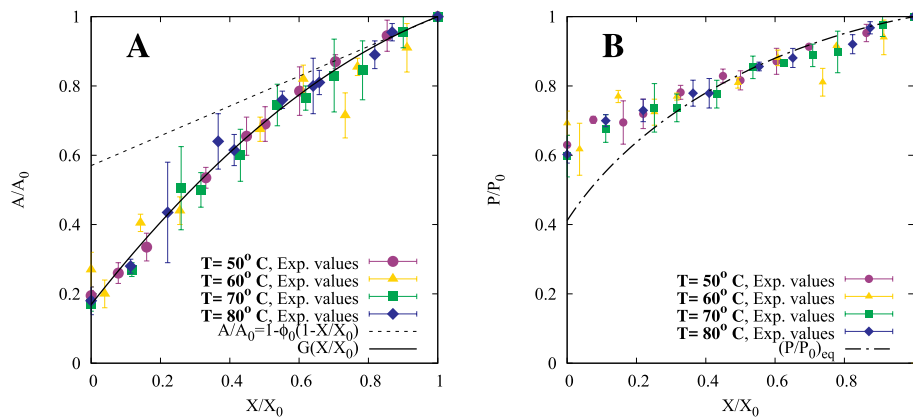


Fig. 5. A) A/A_0 vs X/X_0 for four drying temperatures $T = 50, 60, 70, 80^\circ\text{C}$. Dashed curve represents $A/A_0 = 1 - \phi_0 C(1 - X/X_0)$, Eq. (22) with $C = 1$, i.e. when the volume reduction equals the volume of removed water. Continuous line represents the least-square best fit quadratic curve $G(X/X_0)$, Eq. (29). B) P/P_0 vs X/X_0 for four drying temperatures $T = 50, 60, 70, 80^\circ\text{C}$. Dot-dashed line represents $(P/P_0)_{sq} = (A/A_0)^{1/2} = (G(X/X_0))^{1/2}$ as evaluated from the $G(X/X_0)$ best fit curve, Eq. (29).

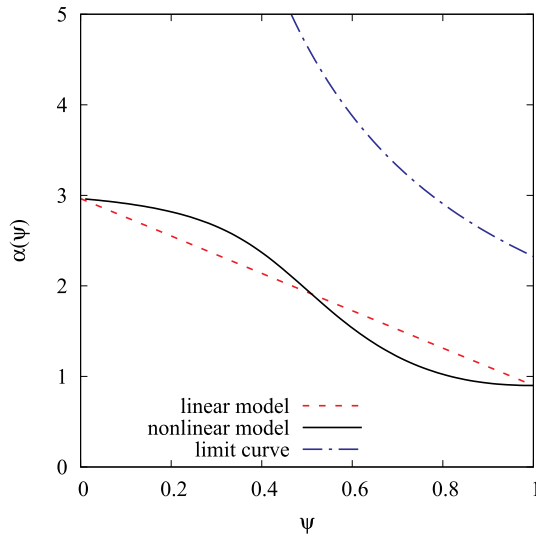


Fig. 6. Shrinkage factor $\alpha(\psi)$ vs ψ for potato slices. Dashed line represents the shrinkage function Eq. (33) obtained from the experimental shrinkage function Eq. (30) with the linear model $X/X_0 = \psi_b$, Eq. (31). Continuous line represents the shrinkage function Eq. (42) obtained from the experimental shrinkage function Eq. (30) with the nonlinear nonlinear model Eq. (36)–(41) with $A_{eq}/A_0 = 0.18$, $Bi_m = 5$, $\phi_0 = 0.43$. Dot-dashed curve represents the limit curve $(\psi\phi_0)^{-1}$.

by the case basis.

Therefore, for the linear decreasing function $\tilde{\alpha}(X/X_0)$ Eq. (30) characterizing potato strips it is expected that, by adopting the linear model Eq. (31), the shrinkage function

$$\alpha(\psi) = \tilde{\alpha}(\psi) = \frac{1}{\phi_0} G'(\psi) \quad (33)$$

will overestimate the real shrinkage factor for large values of X/X_0 while it will underestimate it for smaller values of X/X_0 .

A more refined model for $g(\psi_b, Bi_m)$ can be derived by adopting the following strategy, that develops in the following three steps:

Step 1) To derive the expression for X/X_0 vs ψ_{av}

$$X/X_0 = q(\psi_{av}) \quad (34)$$

by solving Eq. (32) with respect to X/X_0 .

For example, for the quadratic best-fit curve $G(X/X_0)$ for potato slices, Eq. (29), from Eq. (32) one obtains

$$X/X_0 = \psi_{av} G(X/X_0) = \psi_{av} \left[1 - \beta_1 \left(1 - \frac{X}{X_0} \right) - \beta_2 \left(1 - \frac{X}{X_0} \right)^2 \right] \quad (35)$$

By solving the above equation with respect to X/X_0 , it is found the analytic expression relating X/X_0 to ψ_{av}

$$X/X_0 = q(\psi_{av}) = \frac{-b + \sqrt{(b^2 - 4ac)}}{2a}, \quad a(\psi_{av}) = \beta_2 \psi_{av} \quad (36)$$

$$b(\psi_{av}) = 1 - (\beta_1 + 2\beta_2)\psi_{av}, \quad c(\psi_{av}) = (\beta_1 + \beta_2 - 1)\psi_{av}$$

Step 2) To adopt the following general model relating ψ_{av} to ψ_b that takes explicitly into account the influence of Bi_m as follows

$$\psi_{av}(\psi_b; Bi_m) = 1 - \frac{\gamma(Bi_m) \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^2}{1 + \left[\frac{\gamma(Bi_m)}{1 - \psi_{eq}} - 1 - \delta(Bi_m) \right] \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^{1/2} + \delta(Bi_m) \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^2} \quad (37)$$

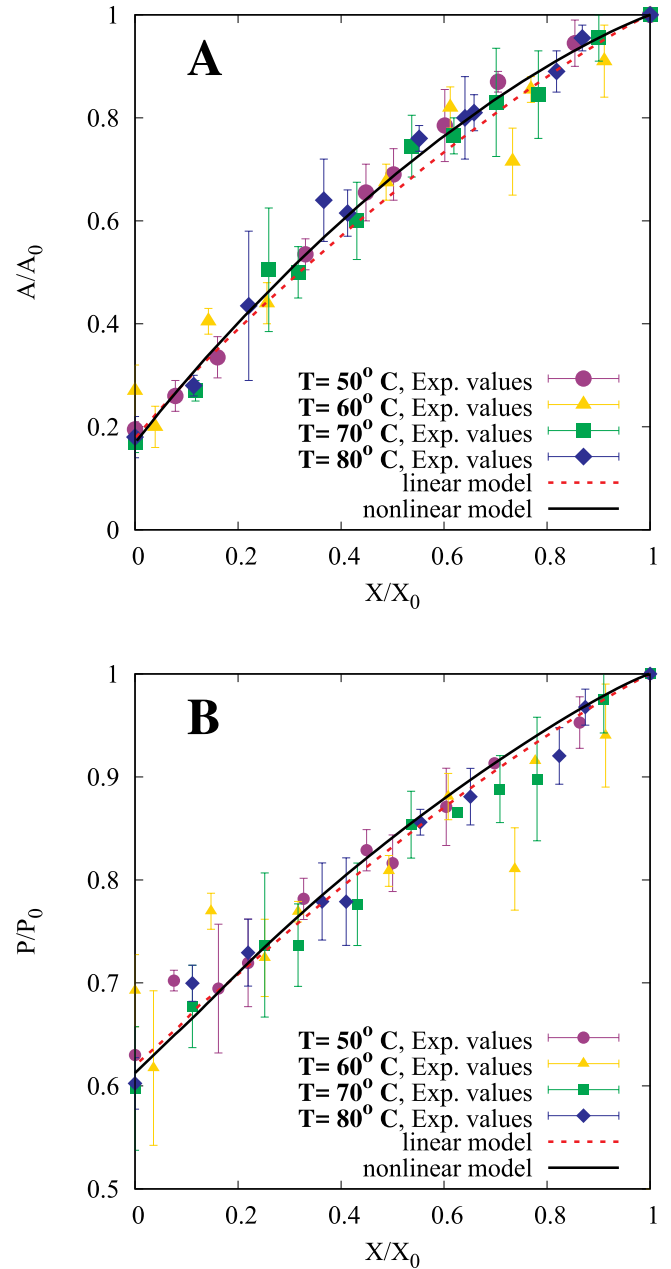


Fig. 7. A) A/A_0 vs X/X_0 for four drying temperatures $T = 50, 60, 70, 80^\circ\text{C}$. B) P/P_0 vs X/X_0 for four drying temperatures $T = 50, 60, 70, 80^\circ\text{C}$. Dashed lines represent model predictions with $\alpha(\psi)$ given by Eq. (33) as obtained from the experimental shrinkage function Eq. (30) with the linear model $X/X_0 = \psi_b$, Eq. (31). Continuous lines represent model predictions with the shrinkage function Eq. (42) obtained from the experimental shrinkage function Eq. (30) with the nonlinear nonlinear model Eq. (36)–(41) with $A_{eq}/A_0 = 0.18$. For both models $Bi_m = 5$, $\phi_0 = 0.43$.

$$\gamma(Bi_m) = \frac{2(2 + Bi_m(1 - \psi_{eq}))}{Bi_m^2} \quad (38)$$

$$\delta(Bi_m) = \frac{(1 - \psi_{eq}) + \gamma(Bi_m)(3 - 2\sigma(Bi_m))}{3(1 - \psi_{eq})} \quad (39)$$

where

$$\sigma(Bi_m) = \frac{Bi_m(A_{eq}/A_0)^{1/2} \sin(\lambda_0)}{2\lambda_0^3} \quad (40)$$

$$2\lambda_0 \tan(\lambda_0) - Bi_m (A_{eq}/A_0)^{1/2} = 0 \quad (41)$$

The symbol λ_0 represents the smallest root of Eq. (41) and A_{eq}/A_0 is the asymptotic rescaled area. The analytic derivation of the equations Eq. (37)–(41) relating ψ_{av} to ψ_b and Bi_m is reported in Appendix I and is independent of the shrinkage function $\alpha(\psi)$.

Step 3) To obtain a more accurate expression for the shrinkage factor

$$\alpha(\psi) = \frac{1}{\phi_0} G'(q(\psi_{av}(\psi_b; Bi_m))) \quad (42)$$

by making use of the analytic expression $X/X_0 = q(\psi_{av})$ Eq. (36) and the analytical expression for $\psi_{av}(\psi_b; Bi_m)$, Eq. (37)–(41).

Fig. 6 shows the comparison between the shrinkage function $\alpha(\psi)$ for potatoes slices obtained from the simplified linear model Eq. (33) and from the nonlinear model Eq. (42). The value $A_{eq}/A_0 \simeq 0.18$ has been directly estimated from experimental data. The value $Bi_m = 5$ has been chosen according to preliminary observations on the perimeter/area ratio. This choice is supported by comparison between model predictions and experimental data shown in Section 5.2.

5.2. Model predictions vs experimental data

Fig. 7 shows the comparison between model predictions and experimental data for A/A_0 vs X/X_0 (Fig. 7 A) and P/P_0 vs X/X_0 (Fig. 7 B) for four drying temperatures $T = 50, 60, 70, 80^\circ\text{C}$. Dotted lines represent the model predictions with the linear model, Eq. (33). Continuous lines represent the model predictions with the nonlinear model, Eq. (42) with $A_{eq}/A_0 \simeq 0.18$ as deduced from experimental data. For both models $Bi_m = 5$. It can be observed that also the simple linear model gives a quite satisfactory prediction of both sample area and perimeter. The nonlinear model captures in a better way the curvature of the experimental curve A/A_0 vs X/X_0 .

It is important to notice that the above analysis is independent of the diffusivity value as the "calibration curve" A/A_0 vs X/X_0 is time independent. The moving boundary model is obviously a time-dependent model usually implemented and numerically integrated with respect to a dimensionless time $\tau = tD/L_r^2$, L_r being a characteristic length of the sample, in the present case the initial side length of the square section $L_r = L_0 = 9.525$ mm.

Once the shrinkage factor $\alpha(\psi)$ has been determined, resulting the same for the four temperatures analyzed - since no significant differences have been observed in the experimental curves A/A_0 vs X/X_0 at the four temperatures - the moving boundary model furnishes a unique

time-dependent curve X/X_0 vs τ and a unique time-dependent curve V/V_0 vs τ for the four temperatures.

The value of the diffusion coefficient $D(T)$ at each drying temperature is determined by means of a least-square best fit of the model curve X/X_0 vs $\tau L^2/D(T)$ onto the experimental data X/X_0 vs t at the corresponding temperature T .

Fig. 8 A shows the experimental data for the time decay of the moisture ratio at four different drying temperatures and the excellent agreement with the nonlinear model predictions. The estimated water diffusivities at different drying temperatures are shown in Fig. 8 B, with the corresponding Arrhenius behaviour $D(T) = D_0 \exp(-E/(RT))$ with $D_0 = 1.00754 \times 10^{-6}$ [m²/s] and $E/R = 2268.62$ [K].

The estimated diffusivities are $D(50^\circ\text{C}) = 9 \times 10^{-10}$ m²/s, $D(60^\circ\text{C}) = 1.1 \times 10^{-9}$ m²/s, $D(70^\circ\text{C}) = 1.383 \times 10^{-9}$ m²/s, $D(80^\circ\text{C}) = 1.616 \times 10^{-9}$ m²/s, in agreement with Hassini et al. (2007) reporting diffusivity values, corrected for product shrinkage, ranging from 3.55×10^{-10} to 1.92×10^{-9} for temperatures varying in the range [40°C–85°C].

Fig. 9 shows the mean edge deformation of slices cut from dried potato strips ($T = 80^\circ\text{C}$) as reported by Ortiz-Garcia-Carrasco et al. (2015) (top) and the corresponding nonlinear model predictions, showing that the shrinkage model proposed is actually able to describe surface deformation and the experimentally observed time-increasing perimeter/area ratio.

6. Conclusions

In this paper a moving-boundary model taking into account volume reduction of food materials during drying, is proposed. The model, borrowed from the literature of swelling of polymer matrices, introduces a pointwise shrinkage velocity. In order to take into account the wide variety of relationships existing between water removal and volume reduction, the pointwise shrinkage velocity is assumed equal and opposite in sign to the water diffusive flux (like in a swelling process) times a pointwise shrinkage factor $\alpha(\psi)$ introduced in order to accurately model different food materials.

Both the theoretical limits of the shrinkage function $\alpha(\psi)$ as well as its influence on the calibration curve V/V_0 vs X/X_0 are investigated. It is shown how a nonlinear behaviour of $\alpha(\psi)$ can lead to large positive or negative curvature of the calibration curve thus making the model capable to capture all the salient features of different food materials subjected to isothermal drying.

A significant part of the work is devoted to the development of a general method for estimating the functional form of the shrinkage factor $\alpha(\psi)$ directly from experimental calibration curves. In this work a

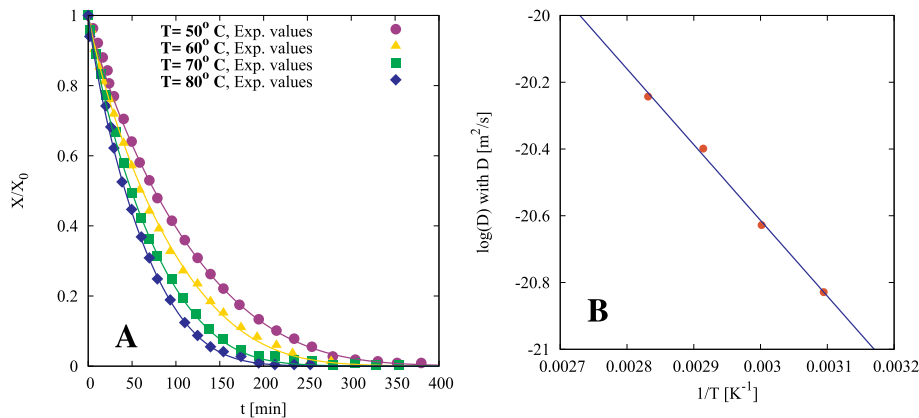


Fig. 8. (A) X/X_0 vs t for four drying temperatures $T = 50, 60, 70, 80^\circ\text{C}$. Continuous lines represent model prediction with the nonlinear model Eqs. (30), (36)–(41), $Bi_m = 5$, $\phi_0 = 0.43$, $A_{eq}/A_0 = 0.18$. B) Water diffusivity $D(T)$ at the drying temperatures $T = 50, 60, 70, 80^\circ\text{C}$. Continuous line represents the Arrhenius behaviour $D(T) = D_0 \exp(-E/(RT))$ with $D_0 = 1.00754 \times 10^{-6}$ [m²/s] and $E/R = 2268.62$ [K].

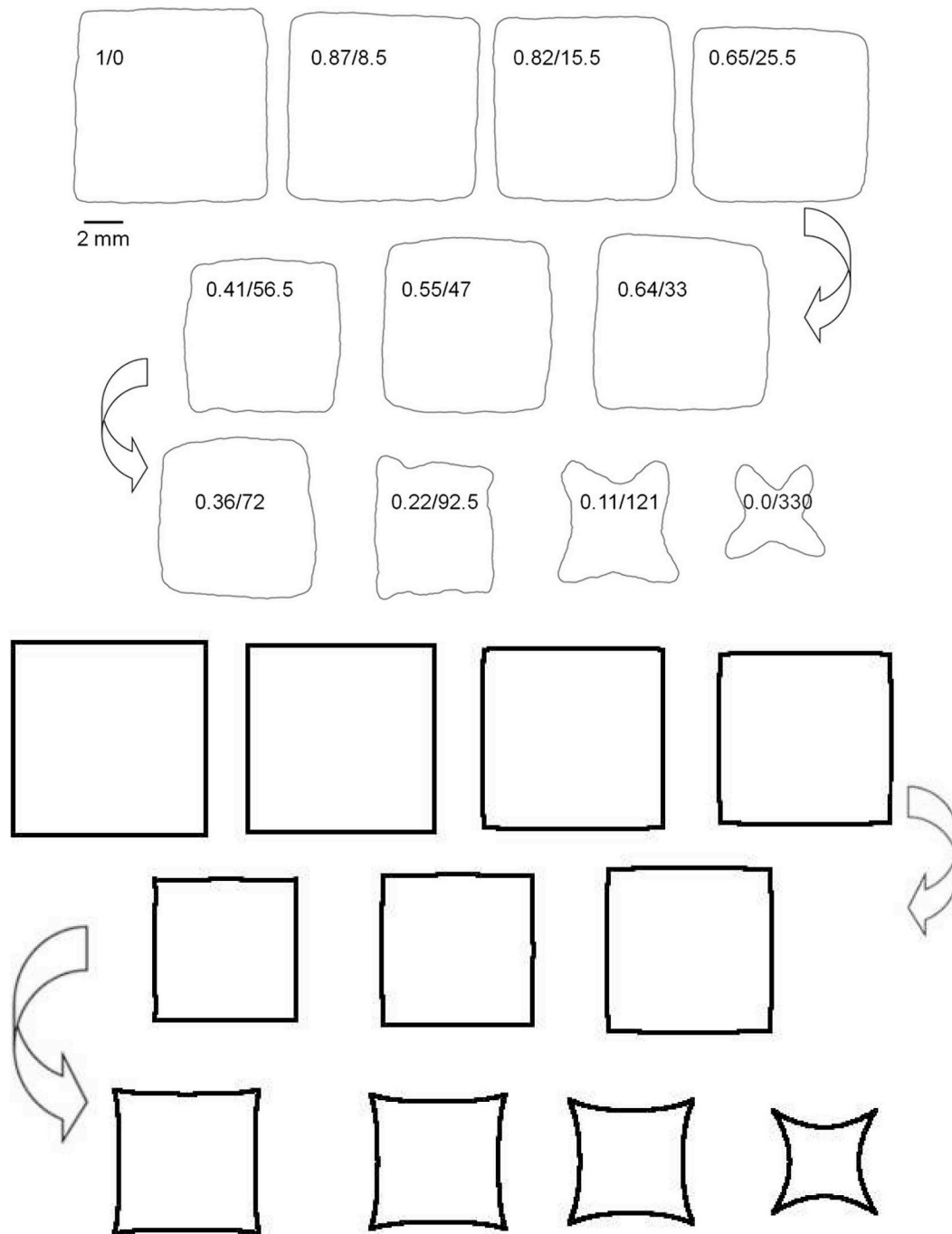


Fig. 9. Top: Mean edge deformation of slices cut from dried potato strips ($T = 80^{\circ}\text{C}$). Inner numbers represent the moisture ratio reached in product/elapsed drying time (min). Bottom: Corresponding nonlinear model predictions at the same time instants for the diffusivity value $D(80^{\circ}\text{C}) = 1.616 \times 10^{-9} \text{ m}^2/\text{s}$.

fully analytical approach is presented, here developed specifically for 2-d square samples (potato slices) but that can be generalized to other geometries.

The role of the mass transfer coefficient, i.e. the Biot number Bi_m , is also investigated. In particular, it is shown that both $\alpha(\psi)$ and Bi_m deeply influence the sample surface deformation during the drying process.

The model is successfully applied to potato strips air-drying, thus furnishing very good results in terms of prediction of volume reduction,

surface deformation and water diffusion coefficient.

In a subsequent work (Adrover et al., 2018) the moving boundary model is adopted for analyzing other literature data of different food materials. The purpose is twofold: (1) to validate the model for different food materials and different samples geometries and (2) to propose a simpler numerical approach for estimating the shrinkage factor directly from experimental calibration curves, thus bypassing too lengthy analytical calculations and developing a general method that can be applied to any sample geometry and any non-linear calibration curve.

Appendix

In this appendix Eq. (37)–(41) are derived. Eq. (37)–(41) relate ψ_{av} to ψ_b at the probe point P_1 , initially placed in $(\xi, \eta) = (0, 1/2)$ for the square 2-d sample with initial dimension $(\xi, \eta) \in [-1/2, 1/2] \times [-1/2, 1/2]$ analyzed in Section 5.

The model Eq. (37)–(41) derives from the analysis of ψ_{av} vs ψ_b in the two limiting cases, of very short and very long (asymptotic) time-scales of the drying process and is independent of the shrinkage function $\alpha(\psi)$.

Short time scales

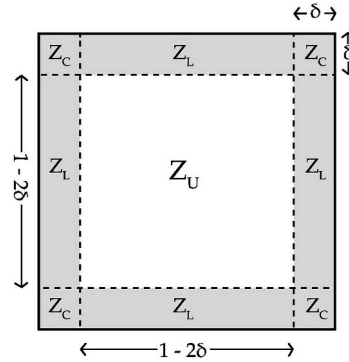


Fig. 10. Schematic representation of the 2-d square domain at the early stage of the drying process. The center part Z_U is the region of uniform water concentration $\psi = 1$. The grey region represents the boundary layer (b.l.) where a linear water concentration profile is assumed.

At the beginning of the drying process, when the sample shrinkage is very small and therefore negligible, it can be reasonably assumed that the water concentration profile is almost uniform in the inner part of the sample (and equal to the initial uniform value $\psi(0) = 1$) and that the concentration gradients are all localized in the proximity of the boundary (boundary layer b.l.) as schematically represented in Fig. 10 where δ is the thickness of the b.l. and the square Z_U , with area $(1 - 2\delta)^2$, is the region of uniform water concentration $\psi = 1$. The boundary layer is decomposed in four identical lateral zones (Z_L) and four identical corner zones (Z_C). Let ψ_b be the water volume fraction on the boundary, assumed uniform on the boundary for sake of simplicity.

By assuming a linear concentration profile in the b.l. the average water concentration in the sample ψ_{av} can be estimated by evaluating the integral of ψ over the central zone Z_U , the four lateral zones Z_L and the four corner zones Z_C , thus obtaining

$$\psi_{av} = \underbrace{\frac{(1 - 2\delta)^2}{\int_{Z_U} \psi \, dx}}_{\int_{Z_U} \psi \, dx} + 4 \underbrace{\left(\delta \frac{(1 - 2\delta)}{2} \frac{1 + \psi_b}{2} \right)}_{\int_{Z_L} \psi \, dx} + 4 \underbrace{\left(\delta^2 \frac{1 - \psi_b}{3} \right)}_{\int_{Z_C} \psi \, dx} \quad (43)$$

The thickness δ of the b.l. depends on Bi_m and can be estimated by enforcing the linear concentration profile in the b.l. and the boundary condition

$$-\frac{\partial \psi}{\partial \xi} \bigg|_{\xi=1/2} = \frac{1 - \psi_b}{\delta} = Bi_m(\psi_b - \psi_{eq}) \rightarrow \delta = \frac{1 - \psi_b}{Bi_m(\psi_b - \psi_{eq})} \quad (44)$$

By substituting Eq. (44) for δ into Eq. (43) for ψ_{av} one obtains an analytic expression for ψ_{av} as a function of ψ_b and Bi_m that is valid at short time scales, i.e. when $\psi_b \simeq 1$. By expanding this expression in power series about the point $\psi_b = 1$ one obtains

$$\psi_{av} = 1 - \frac{2(2 + Bi_m(1 - \psi_{eq}))}{Bi_m^2} \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^2 + \mathcal{O}(1 - \psi_b)^3 \quad (45)$$

and therefore a quadratic behaviour is predicted for $1 - \psi_{av}$ vs $1 - \psi_b$ at the early stage of the drying process, i.e. for $\psi_b \simeq 1$.

Long time scales

At long time scales, when the sample shrinkage has almost reached its asymptotic value and concentration gradients are very small, the contribution of the shrinkage velocity $\mathbf{v}$ (proportional to the pointwise concentration gradient) in the material balance Eq. (3) is negligible and therefore the water concentration profile $\psi(\xi, \eta, \tau)$ can be evaluated by solving a pure diffusion problem in a 2-d fixed boundary domain. The asymptotic rescaled sample area is A_{eq}/A_0 and the corresponding sample length can be approximated as $\tilde{L}_{eq} = L_{eq}/L_0 = \sqrt{A_{eq}/A_0}$.

By introducing $\tilde{\psi} = \psi - \psi_{eq}$ defined in the dimensionless domain $(\xi, \eta) \in \left(\frac{-\tilde{L}_{eq}}{2}, \frac{\tilde{L}_{eq}}{2} \right) \times \left(\frac{-\tilde{L}_{eq}}{2}, \frac{\tilde{L}_{eq}}{2} \right)$ and satisfying a pure diffusion equation with symmetry and homogeneous Robin boundary conditions

$$\frac{\partial \tilde{\psi}}{\partial \tau} = \frac{\partial^2 \tilde{\psi}}{\partial \xi^2} + \frac{\partial^2 \tilde{\psi}}{\partial \eta^2}, \quad (\xi, \eta) \in \left(\frac{-\tilde{L}_{eq}}{2}, \frac{\tilde{L}_{eq}}{2} \right) \times \left(\frac{-\tilde{L}_{eq}}{2}, \frac{\tilde{L}_{eq}}{2} \right) \quad (46)$$

$$\frac{\partial \tilde{\psi}}{\partial \xi} \bigg|_{\xi=0} = 0, \quad \frac{\partial \tilde{\psi}}{\partial \xi} \bigg|_{\xi=\tilde{L}_{eq}/2} + Bi_m \tilde{\psi}|_{\xi=\tilde{L}_{eq}/2} = 0 \quad (47)$$

$$\left. \frac{\partial \tilde{\psi}}{\partial \eta} \right|_{\eta=0} = 0, \quad \left. \frac{\partial \tilde{\psi}}{\partial \eta} \right|_{\eta=\tilde{L}_{eq}/2} + Bi_m \tilde{\psi}|_{\eta=\tilde{L}_{eq}/2} = 0 \quad (48)$$

one obtains

$$\tilde{\psi}(\xi, \eta, \tau) = \sum_{h=0}^{\infty} \sum_{k=0}^{\infty} a_{h,k}(\tau) \cos\left(\frac{2\lambda_h \xi}{\tilde{L}_{eq}}\right) \cos\left(\frac{2\lambda_k \eta}{\tilde{L}_{eq}}\right) \quad (49)$$

$$a_{h,k}(\tau) = a_{h,k}(0) \exp[-(\lambda_h^2 + \lambda_k^2)\tau] \quad (50)$$

$$2\lambda_j \tan(\lambda_j) - Bi_m \tilde{L}_{eq} = 0, \quad j = 0, 1, \dots, \infty \quad (51)$$

and the quantities

$$\tilde{\psi}_{av}(\tau) = \frac{4}{\tilde{L}_{eq}^2} \int_0^{\tilde{L}_{eq}/2} \int_0^{\tilde{L}_{eq}/2} \tilde{\psi} d\xi d\eta = \sum_{h=0}^{\infty} \sum_{k=0}^{\infty} a_{h,k}(\tau) \frac{\sin(\lambda_h) \sin(\lambda_k)}{\lambda_h \lambda_k} \quad (52)$$

$$\tilde{\psi}_b(\tau) = \tilde{\psi}\left(0, \frac{\tilde{L}_{eq}}{2}, \tau\right) = \sum_{h=0}^{\infty} \sum_{k=0}^{\infty} a_{h,k}(\tau) \cos(\lambda_k) \quad (53)$$

where $(\xi, \eta) = \left(0, \frac{\tilde{L}_{eq}}{2}\right)$ represents the asymptotic position of the probe point P_1 initially placed in $(\xi, \eta) = (0, 1/2)$.

In the asymptotic limit $\tau \rightarrow \infty$, given the exponential decay of $a_{h,k}(\tau)$, the leading term is $a_{0,0}(\tau) = a_{0,0} \exp[-2\lambda_0^2 \tau]$ associated with the dominant eigenvalue λ_0 , given by the smallest positive root of Eq. (51), thus obtaining

$$\lim_{\tau \rightarrow \infty} \tilde{\psi}_{av}(\tau) = a_{0,0}(\tau) \frac{[\sin(\lambda_0)]^2}{\lambda_0^2}, \quad \lim_{\tau \rightarrow \infty} \tilde{\psi}_b(\tau) = a_{0,0}(\tau) \cos(\lambda_0) \quad (54)$$

By merging together Eq. (54) the following relation between $\tilde{\psi}_{av}$ and $\tilde{\psi}_b$ is obtained, valid only for long time scales

$$\tilde{\psi}_{av} = \left(\frac{\sin(\lambda_0)}{\lambda_0^3} \frac{Bi_m}{2} \tilde{L}_{eq} \right) \tilde{\psi}_b = \underbrace{\left(\frac{\sin(\lambda_0)}{\lambda_0^3} \frac{Bi_m}{2} \sqrt{\frac{A_{eq}}{A_0}} \right)}_{\sigma(Bi_m)} \tilde{\psi}_b \quad (55)$$

Eq. (55), rewritten in terms of the original variables ψ_{av} and ψ_b reads as

$$\psi_{av} = \psi_{eq} + \sigma(Bi_m)(\psi_b - \psi_{eq}) \quad (56)$$

highlighting a linear relationship between ψ_{av} and $(\psi_b - \psi_{eq})$ at long time scales of the drying process, i.e. for $\psi_b \simeq \psi_{eq}$

From short to long time scales

In order to have an expression for ψ_{av} vs ψ_b that is valid in the entire range of values $\psi_b \in (\psi_{eq}, 1)$ the sigmoidal function Eq. (37) is adopted

$$\psi_{av}(\psi_b, Bi_m) = 1 - \frac{\gamma(Bi_m) \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^2}{1 + \left[\frac{\gamma(Bi_m)}{1 - \psi_{eq}} - 1 - \delta(Bi_m) \right] \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^{1/2} + \delta(Bi_m) \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^2} \quad (57)$$

Eq. (57) satisfies the two constraints $\psi_{av}(1) = 1$ (uniform initial water concentration) and $\psi_{av}(\psi_b = \psi_{eq}) = \psi_{eq}$ (uniform water concentration equal to the equilibrium concentration ψ_{eq} at the end of the drying process).

The two parameters $\gamma(Bi_m)$ and $\delta(Bi_m)$ entering Eq. (57) can be obtained by enforcing the two limit behaviours, Eq. (45) for $\psi_b \simeq 1$ and Eq. (56) for $\psi_b \simeq \psi_{eq}$.

By expanding Eq. (57) in power series about the point $\psi_b = 1$ one obtains

$$\psi_{av} = 1 - \gamma(Bi_m) \left(\frac{1 - \psi_b}{1 - \psi_{eq}} \right)^2 + \mathcal{O}(1 - \psi_b)^{5/2} \quad (58)$$

By comparing Eq. (58) with Eq. (45) valid at the early stage of the drying process, i.e. when $\psi_b \simeq 1$, one obtains the following expression for $\gamma(Bi_m)$

$$\gamma(Bi_m) = \frac{2(2 + Bi_m(1 - \psi_{eq}))}{Bi_m^2} \quad (59)$$

By expanding Eq. (57) in power series about the point $\psi_b = \psi_{eq}$ one obtains

$$\psi_{av} = \psi_{eq} + \frac{(1 - \psi_{eq}) + 3\gamma(Bi_m) - 3\delta(Bi_m)(1 - \psi_{eq})}{2\gamma(Bi_m)} (\psi_b - \psi_{eq}) \quad (60)$$

By comparing Eq. (60) with the asymptotic behaviour Eq. (56) one obtains the following expression for $\delta(Bi_m)$

$$\delta(Bi_m) = \frac{(1 - \psi_{eq}) + \gamma(Bi_m)(3 - 2\sigma(Bi_m))}{3(1 - \psi_{eq})} \quad (61)$$

where $\sigma(Bi_m)$ is given by Eq. (55).

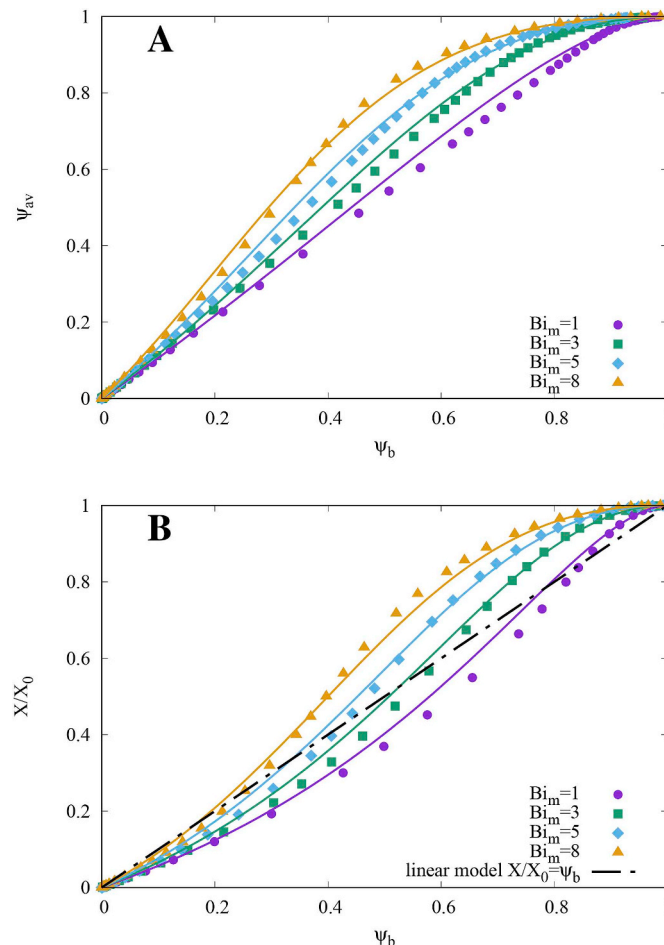


Fig. 11. Numerical results for ψ_{av} vs ψ_b (A) and X/X_0 vs ψ_b (B) at the probe point P_1 initially placed at $(\xi, \eta) = (0, 1/2)$. $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$, $Bi_m = 1, 3, 5, 8$. Continuous lines represent the nonlinear model Eqs. ((57), (59), (61) and (62).

It is important to stress that the analytical derivation proposed above is independent of the shrinkage function $\alpha(\psi)$ and only the asymptotic area ratio A_{eq}/A_0 enters into Eqs. (57), (59) and (61) that can be therefore applied, in principle, to any food material.

The analytic derivation can be easily generalized to rectangular cross-sections as well as to other two-dimensional geometries, e.g. cylindrical or discoid samples.

Reliability and accuracy of the analytical expression Eqs. (57), (59) and (61) are confirmed by direct comparison with numerical results for ψ_{av} vs ψ_b , shown in Fig. 11 A, obtained for $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$, i.e. the ideal shrinkage benchmark case for $Bi_m = 1, 3, 5, 8$. Fig. 11 B shows the good agreement between numerical data for X/X_0 vs ψ_b and the model predictions in the ideal shrinkage benchmark case for which, from Eq. (32) and from the relation $A/A_0 = 1 - C\phi_0(1 - X/X_0)$ one obtains

$$X/X_0 = \frac{\psi_{av}(1 - C\phi_0)}{1 - \psi_{av}C\phi_0} \quad (62)$$

The agreement between numerical data and the analytic expressions is quite satisfactory if one considers that the proposed expression for $\psi_{av}(\psi_b; Bi_m)$ has no fitting parameters and is reliable in a wide range of Bi_m values.

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